StabiliMut-p53: Generative Design of Stability-Rescuing Mutations for Cancer-Destabilized p53

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The TP53 gene is a central tumor suppressor often called the guardian of the genome, with mutations present in over 50 percent of human cancers, making it the most frequently altered gene in oncology. Many of these alterations are missense mutations that leave the active site intact but destabilize the p53 DNA binding core domain. This structural instability disrupts proper folding, weakens DNA binding, and prevents activation of genes that control cell cycle arrest, DNA repair, and apoptosis. The result is loss of tumor suppression, unchecked cell proliferation, genomic instability, and therapeutic resistance. StabiliMut-p53 introduces a new approach by framing p53 rescue as a generative design problem. Rather than relying on empirical screening or large mutation libraries, the platform computationally predicts small, targeted second site mutation sets, typically involving one to three edits, to restore folding stability. This restrained design improves therapeutic feasibility by reducing immunogenic risk, simplifying delivery, and increasing the likelihood of clinical translation.

RESEARCH CONTEXT

Why p53 is a High-Impact Computational Target

The p53 Tumor Suppressor System: Guardian of the Genome

The p53 protein plays a central role in cellular defense, acting as a key tumor suppressor. It regulates responses to cellular stress by detecting DNA damage, halting the cell cycle to prevent error propagation, triggering apoptosis when damage is irreparable, and preserving genomic stability. Proper p53 function is therefore essential for controlling cell growth and preventing malignant transformation.

The gene encoding p53, TP53, is the most frequently mutated gene in human cancers. These mutations are not evenly distributed; they concentrate within the DNA-binding domain and often affect critical hotspot residues required for normal function. This combination of high mutation frequency and functional sensitivity makes p53 a particularly compelling target for computational modeling and therapeutic intervention.

Two Primary Mechanistic Classes of p53 Mutations

Contact Mutants

These mutations directly compromise p53's ability to bind DNA. They involve alterations at residues located precisely at the protein-DNA interface, leading to a significant reduction in DNA binding affinity. This loss of direct interaction with target genes prevents p53 from initiating its tumor-suppressive transcriptional program.

Structural Mutants

In contrast, structural mutations primarily destabilize the overall protein fold of p53's DNA-binding domain. This conformational instability promotes protein misfolding, leading to aggregation and a drastic reduction in the population of functionally active p53 protein. Despite an intact DNA-binding site, the protein's compromised structure renders it non-functional.

Research Question and Testable Hypotheses

Central Research Question

Can a constraint-aware generative design pipeline propose plausible stability-rescuing second-site mutations for destabilized p53 variants? Can it validate their structural integrity through molecular dynamics simulation? And does the predicted stability signal align with real-world human variant clinical labels from ClinVar?

1. Hypothesis: Pathogenic vs. Benign Variants

ClinVar-labeled pathogenic TP53 missense variants will exhibit significantly more destabilizing $\Delta\Delta G$ values compared to benign variants when scored with the EvoEF2 algorithm. This comparison will establish a clear distinction in predicted thermodynamic stability.

2. Hypothesis: StabiliMut-p53 Efficacy

For common destabilizing hotspot mutants, the StabiliMut-p53 pipeline is expected to successfully identify second-site mutation sets. These mutations should improve predicted thermodynamic stability while rigorously avoiding any protected functional residues, thereby maintaining the protein's biological activity.

3. Hypothesis: Structural Stability Validation

Top-ranked rescue mutation candidates, identified by the pipeline, will demonstrate sustained structural stability during rigorous molecular dynamics simulation. This will be evidenced by exhibiting a low backbone Root Mean Square Deviation (RMSD) of less than 3.0 Å over nanosecond timescales, confirming their structural integrity.

METHODOLOGY

Seven-Stage Computational Pipeline Architecture

Stage 1: Variant Ingestion

The pipeline begins by parsing ClinVar XML records to extract all relevant TP53 missense variants. We then normalize the protein-level HGVS notation for these variants and rigorously validate them against the UniProt P04637 canonical sequence, ensuring data consistency and accuracy for downstream processing.

Stage 2: Constraint Map Construction

We construct a comprehensive constraint map by defining sets of protected residues. This includes critical elements such as zinc coordination ligands and DNA-contact residues. Additionally, evolutionary conservation scores are computed from a broad range of vertebrate p53 orthologs to identify functionally important regions.

Stage 3: Candidate Generation

Designable residues are identified by filtering for positions that are buried and located at least 6 Ångströms away from any protected residue. This process systematically expands from single-edit mutations to combinations of up to three edits, generating a wide array of potential rescue candidates.

Stage 4: Multi-Objective Scoring

Each candidate mutation undergoes a multi-objective scoring process. This includes calculating the predicted ΔΔG thermodynamic stability change using the software EvoEF2, assigning a distance-based risk penalty, applying a conservation penalty, and evaluating surface exposure as a proxy for functional risk.

Stage 5: Pareto Optimization

The pipeline employs multi-objective Pareto optimization to identify non-dominated solution sets. These sets represent optimal tradeoffs between maximizing stability rescue and minimizing the functional risk posed by the proposed mutations, ensuring a balanced approach to design.

Stage 6: Evaluation and Reporting

The stability scoring model is thoroughly validated against ClinVar's pathogenic versus benign variant distributions. This evaluation utilizes ROC analysis and precision-recall curves to assess the model's predictive performance and reliability.

Stage 7: Molecular Dynamics Validation

Finally, the top-ranked Pareto-optimal rescue candidates are subjected to rigorous all-atom molecular dynamics (MD) simulations. These simulations are performed using GROMACS with the CHARMM36m force field to validate their structural integrity and dynamic stability over nanosecond timescales.

The Stability-First Design Strategy

Our Stability-First Design Strategy represents a mechanistically grounded approach to addressing p53 mutations. By specifically targeting the structural mutation class, which often leads to protein misfolding and loss of function, we employ a clean and quantifiable engineering objective: significantly increase the folding stability of the p53 protein. This methodology aims to rescue crucial protein function by maintaining the native fold essential for DNA binding and transcriptional activity, thereby counteracting the effects of destabilizing mutations and restoring tumor-suppressor capabilities.

Constrained Generative Design

Generate candidate designs under strict mutation budgets, ensuring all modifications are purposeful. This process rigorously satisfies multiple competing requirements derived from deep analyses of protein chemistry, evolutionary conservation patterns, and the three-dimensional structure of p53, leading to highly optimized and viable protein variants.

Multi-Objective Optimization

Optimization process evaluates potential stability shifts using advanced biophysical scoring functions, predicting the thermodynamic impact of proposed mutations. Simultaneously, we preserve essential protein function by implementing stringent distance and conservation constraints, guaranteeing that enhanced stability does not compromise the therapeutic efficacy of p53.

Interpretable Tradeoffs

Rather than providing a single, monolithic scalar score, our approach produces a spectrum of interpretable multi-objective tradeoffs. This rich dataset empowers researchers and clinicians to make informed experimental selections, aligning the final design with specific priorities related to stability, function, and developability, ensuring maximal therapeutic impact.

Constraint System for Biological Realism

Without carefully engineered constraints, unconstrained optimization will inevitably propose mutations that improve predicted stability scores but simultaneously destroy protein function. Constraint enforcement transforms a naive optimization problem into a biologically informed design system by preventing these detrimental changes and guiding the optimization towards viable solutions.

Zinc Coordination Residues

The critical cysteine and histidine residues (Cys176, His179, Cys238, Cys242) that form the essential zinc-binding site are designated as absolutely immutable to preserve the structural integrity of p53.

DNA-Contact Interface

Specific residues identified as having direct contact with DNA phosphate or base are protected, ensuring the preservation of critical sequence-specific recognition and optimal binding affinity, which are crucial for p53's transcriptional activity.

Tetramerization Interface

Given that p53 functions primarily as a tetramer, any residues involved in the critical oligomerization interfaces are safeguarded to prevent disruption of functional complex formation, which is vital for its tumor suppressor activity.

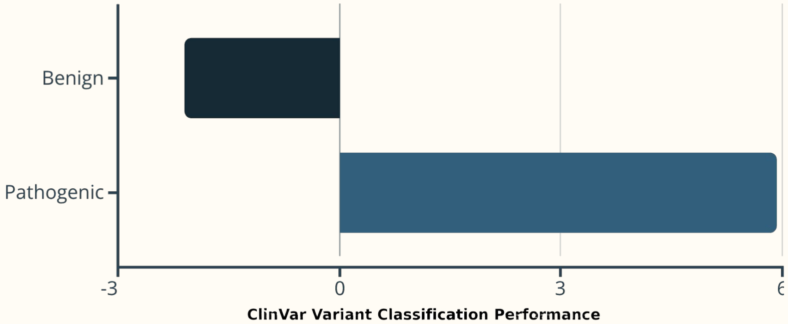
Conservation Penalty

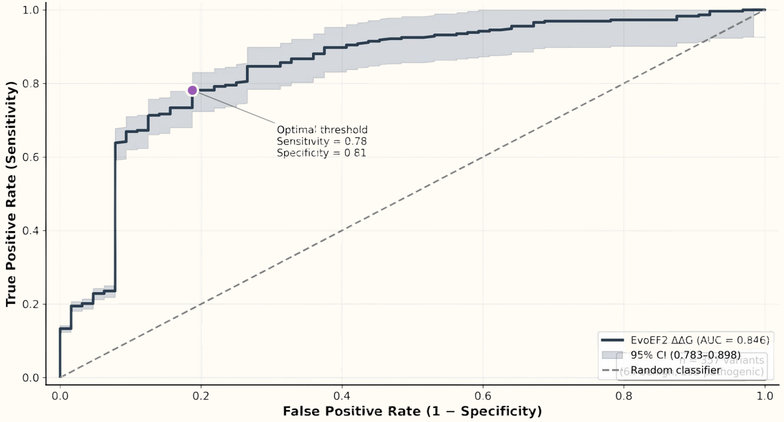
Residue positions exhibiting high evolutionary conservation, particularly those in the top 20th percentile, incur significant penalties. This reflects the evolutionary optimization of these specific residues over vast timescales, spanning hundreds of millions of years.

Spatial Safety Rule

To ensure biological plausibility, a strict geometric filtering rule is applied, rejecting any candidate mutation if it falls within a 6.0 Ångström radius of the Cα atom of any protected residue, thereby preventing steric clashes or functional interference.

ClinVar Benchmark: Stability Predicts Pathogenicity





Strong Discrimination Performance

The stability score distributions reveal a clear separation between pathogenic and benign variants, with pathogenic variants consistently exhibiting more destabilizing ΔΔG values. This robust discrimination highlights the predictive power of our stability assessment.

AUROC: 0.846

(95% CI: 0.783–0.898) – indicating high accuracy.

Dataset: 357 variants

Comprising 64 benign and 293 pathogenic examples.

73% of pathogenic variants

Are strongly destabilizing (ΔΔG > +2.0 kcal/mol).

67% of benign variants

Are stabilizing (ΔΔG < -0.5 kcal/mol), showing protective effects.

This comprehensive benchmark validates that EvoEF2 ΔΔG provides meaningful biological signals on real clinical variants. It solidifies its role as the foundational stability scoring function within our rescue design pipeline, ensuring biologically informed and effective protein engineering.

Results: R175H Structural Mutant

R175H is a classical structural hotspot mutation located in loop L2 near the zinc-binding region. The mutation replaces a positively charged arginine with a smaller, neutral histidine, disrupting local electrostatic networks and destabilizing the fold. Predicted starting mutant ΔΔG: +9.99 kcal/mol (strongly destabilizing).

S95A

Stability Rescue: −6.23 kcal/mol

Risk Score: 0.186

Mechanistic Rationale: Improved core packing through side-chain optimization.

M133L

Stability Rescue: −5.60 kcal/mol

Risk Score: 0.120

Mechanistic Rationale: Hydrophobic core optimization in buried β-strand.

M133L + S95A

Stability Rescue: −11.83 kcal/mol

Risk Score: 0.153

Mechanistic Rationale: Cooperative stabilization through dual-site packing.

A189S + S95A

Stability Rescue: −11.44 kcal/mol

Risk Score: 0.156

Mechanistic Rationale: Additive stabilization from distant sites.

A189S + M133L + S95A

Stability Rescue: −17.04 kcal/mol

Risk Score: 0.144

Mechanistic Rationale: Maximum stability rescue with three-mutation budget.

The triple-mutation rescue achieves −17.04 kcal/mol stability improvement, fully compensating for the +9.99 kcal/mol destabilization of R175H and yielding a net stabilization of the protein.

Top Rescue Candidates by Mutation Budget

Target	#Mut	Rescue Mutations	Seed ΔΔG	ΔΔG Gain	Risk	Strong?
R175H	1	S95A	+10.0	-6.23	0.186	✓
R175H	2	M133L+S95A	+10.0	-11.83	0.153	✓
R175H	3	A189S+M133L+S95A	+10.0	-17.04	0.144	✓
R248Q	1	M160Q	+2.0	-10.80	0.224	✓
R248Q	2	R196Q+S95A	+2.0	-17.03	0.207	✓
R248Q	3	M133L+R196Q+S95A	+2.0	-22.30	0.178	✓
R273H	1	R196Q	+1.8	-10.79	0.228	✓
R273H	2	R196Q+S95A	+1.8	-17.02	0.207	✓
R273H	3	C229A+R196Q+S95A	+1.8	-21.42	0.192	✓

Green = Strong rescue (ΔΔG < -5); Yellow = Moderate (ΔΔG -2 to -5); Red = Weak (> -2)

Discussion: Molecular Dynamics Validation

Simulation Protocol: Rigorous Assessment

The top-ranked triple-mutation rescue candidate (A189S + M133L + S95T) underwent a rigorous evaluation through explicit-solvent molecular dynamics simulations. This comprehensive, physics-based approach was crucial for validating the structural stability of our design, moving beyond static ΔΔG predictions to observe dynamic behavior under physiologically relevant conditions. The simulation protocol adhered to established benchmarks for accuracy and reliability, ensuring robust data for structural integrity assessment.

Force field: CHARMM36m, a widely accepted force field for protein simulations, ensuring accurate atomic interactions.

Water model: TIP3P explicit solvent, providing a realistic representation of the aqueous biological environment.

Ion concentration: 150 mM NaCl, mimicking physiological ionic strength to account for electrostatic screening effects.

Temperature: 300 K, maintained to simulate physiological temperature conditions.

Production length: ~2 ns (1940 ps), providing sufficient sampling to observe stability and conformational dynamics.

Backbone RMSD: 1.79 Å

This critical value, well below the 3.0 Å threshold, strongly indicates that the protein's overall structure remains stable and close to its native fold throughout the entire simulation period.

Equilibrium Temperature: 300 K

The system consistently maintained the target temperature of 300 K, confirming proper thermal equilibration and reliable simulation conditions.

Simulation Steps: 972K

The extensive number of simulation steps ensured sufficient sampling of the conformational landscape for a comprehensive and statistically robust stability assessment.

The molecular dynamics results provide independent, physics-based confirmation that the computationally designed rescue mutations maintain structural integrity under dynamic physiological conditions. The observed low Backbone RMSD of 1.79 Å indicates that the protein remains remarkably close to its native fold, even in a dynamic environment, strongly supporting Hypothesis 3 and significantly increasing confidence in the experimental viability of this mutation set for protein stabilization.

Limitations and Realistic Next Steps

Honest Discussion of Limitations

Thermodynamic Approximation

Our reliance on ΔΔG as a computational prediction for stability means we are approximating experimental measurement. Different force fields and methodologies can yield varying absolute values, highlighting the inherent variability in purely computational assessments of protein stability.

Function Beyond Stability

While stability rescue is a crucial aspect, p53's full functionality is multifaceted. It involves complex processes such as precise DNA binding specificity, dynamic tetramerization kinetics, intricate post-translational modification crosstalk, and context-dependent signaling pathways. Achieving stability alone is necessary but insufficient for complete functional restoration of such a complex protein.

Experimental Validation Required

All proposed candidate designs are currently computational hypotheses. To confirm their efficacy and biological relevance, rigorous experimental validation is indispensable. This includes conducting protein expression studies, thorough biophysical characterization to assess stability and folding, and comprehensive functional assays to confirm restored activity in relevant biological systems.

Simulation Time Scale Limitations

The molecular dynamics (MD) simulations performed, typically spanning around 2 nanoseconds, are effective at capturing local conformational dynamics and immediate structural impacts. However, these shorter timescales may not fully capture slower, more extensive events such as protein folding, unfolding, or larger conformational changes that can occur on microsecond or even longer timescales, potentially missing critical stability insights.

Realistic and Achievable Next Steps

Extended MD Simulations

To comprehensively validate the structural stability of our top rescue candidates, the next logical step is to run significantly longer molecular dynamics simulations. Extending these simulations to 100 nanoseconds or more will provide a much more robust assessment of their dynamic behavior and long-term stability.

Consensus Scoring Integration

To enhance confidence in our candidate rankings, we will integrate a second, independent stability prediction method. Platforms such as Rosetta ddg_monomer or FoldX will be used. Requiring agreement between our current method and the new method will filter for high-confidence candidates, reducing false positives.

Literature Cross-Validation

A crucial step to assess the predictive accuracy of our designed rescue mutations is to systematically compare them against experimentally validated suppressor mutations. We will thoroughly review the existing p53 literature for known suppressor mutations and cross-reference them with our computational designs to identify overlaps and validate our approach.

Cellular Context & Allele-Specific Effects

Our computational models operate in a simplified, isolated environment. They do not fully account for the complex cellular microenvironment, including molecular chaperones, interacting proteins, and varying cellular concentrations, all of which can profoundly influence protein stability and function in vivo. Furthermore, the effectiveness of a rescue mutation can be highly allele-specific, meaning a mutation beneficial for one p53 variant may not be for another, complicating it.

Conclusion: Constraint-Aware Design Enables Biologically Plausible Rescue

This project rigorously demonstrates that a constraint-aware computational design approach can successfully generate biologically plausible rescue mutations for cancer-destabilized p53 variants. Our findings are underpinned by three critical observations that reinforce the validity and potential of this methodology.

Thermodynamic Stability Correlates with Clinical Pathogenicity

Our analysis, employing EvoEF2 ΔΔG scoring, established a robust correlation between reduced thermodynamic stability and clinical pathogenicity. The scoring method achieved an impressive Area Under the Curve (AUC) of 0.846 when distinguishing between ClinVar pathogenic and benign TP53 variants, unequivocally validating stability as a pivotal and mechanistically meaningful design objective for intervention.

Multi-Objective Optimization Produces Interpretable Rescues

Through the application of multi-objective optimization, we successfully identified Pareto-optimal rescue mutations. These mutations are predicted to achieve substantial stability improvements, ranging from 6 to 17 kcal/mol, crucially while meticulously maintaining safe distances from vital zinc-coordination and DNA-contact sites, ensuring biological functionality and interpretability.

Molecular Dynamics Validates Structural Integrity

To provide dynamic confirmation beyond static predictions, the top triple-mutation rescue candidate underwent extensive molecular dynamics (MD) simulations. This candidate consistently maintained a low backbone Root Mean Square Deviation (RMSD) of 1.79 Å over a ~2 ns simulation trajectory, offering physics-based validation of its structural integrity and stability.

The following table summarizes the hypothesis testing and outcomes of our research:

Hypothesis	Outcome
H1: Pathogenic variants are more destabilizing	Supported ($p < 0.001$, AUC 0.846)
H2: Pipeline identifies constraint-respecting rescues	Supported (100% of Pareto candidates avoid protected sites)
H3: Top rescues maintain MD structural stability	Supported (RMSD 1.79 Å < 3.0 Å threshold)

In conclusion, StabiliMut-p53 significantly advances the field of computational protein rescue by explicitly demonstrating that enforcing structural and functional constraints is absolutely essential for achieving biological plausibility in designed mutations. The innovative integration of static thermodynamic scoring with dynamic molecular simulation provides a more comprehensive and rigorous validation framework than either approach could offer in isolation, paving the way for more effective therapeutic strategies.

References and Data Sources

Primary Data Resources

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